

Network Intrusion Detection System (NIDS) Using Snort

Task 4 Report — Snort 3 Configuration, Rule Creation & Traffic Monitoring

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Task: CodeAlpha Cyber Security Internship — Task 4
Project: Network Intrusion Detection System Using Snort

Tools: Snort 3 | Operating System: Kali Linux | Interface: eth0

1. Introduction

A Network Intrusion Detection System (NIDS) is a security system used to monitor network traffic and detect suspicious or malicious activities. In this task, Snort 3 was configured as a network-based intrusion detection system to analyze traffic and generate alerts based on predefined security rules.

2. Objective

The objectives of this task were:

- Install and configure Snort as a Network Intrusion Detection System.
- Create custom detection rules.
- Monitor network traffic continuously.
- Generate alerts for suspicious activities.
- Store and analyze intrusion events.

3. Tools and Environment

Tool	Snort 3
Operating System	Kali Linux
Network Interface	eth0

4. Installation and Configuration

Snort was installed on Kali Linux and its configuration was validated successfully.

Configuration testing command:

```
sudo snort -T -c /etc/snort/snort.lua
```

Result:

```
Snort successfully validated the configuration (with 0 warnings).
```

5. Custom Detection Rules

Custom rules were created in the `local.rules` file to detect different types of network activities.

Implemented rules:

ICMP Ping Detection

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

HTTP Traffic Detection

```
alert tcp any any -> any 80 (msg:"HTTP Traffic Detected"; sid:1000002; rev:1;)
```

HTTPS Traffic Detection

```
alert tcp any any -> any 443 (msg:"HTTPS Traffic Detected"; sid:1000003; rev:1;)
```

6. Network Monitoring

Snort was executed in live monitoring mode to analyze network traffic.

Command used:

```
sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/local.rules \
-i eth0 -A alert_fast
```

Snort successfully started monitoring the network interface and captured live traffic.

7. Alert Detection and Testing

Testing was performed by generating ICMP traffic using the ping command:

```
ping -c 4 8.8.8.8
```

Snort successfully detected the activity and generated alerts:

```
[1:1000001:1] "ICMP Ping Detected"
{ICMP} 10.0.2.15 -> 8.8.8.8
```

This confirmed that the intrusion detection rules were working correctly.

8. Response Mechanism

Snort provided real-time alerts whenever suspicious traffic matched the configured rules. These alerts can be used by security teams for further investigation and response actions.

9. Conclusion

In this task, Snort 3 was successfully configured as a Network Intrusion Detection System. Custom rules were created to detect ICMP, HTTP, and HTTPS traffic. Real-time monitoring and alert generation were tested successfully, demonstrating the ability of Snort to identify suspicious network activities.

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